# 1 Quick refresher on the current in electrodynamics

## 1.1 Maxwell's equations

Given Maxwell's equations,

$$\vec{\nabla}\cdot\vec{E}(\vec{r},t) = \frac{1}{\varepsilon_0}\,\rho(\vec{r},t) \qquad \text{(A.1a)} \qquad (1)$$

$$\vec{\nabla}\cdot\vec{B}(\vec{r},t) = 0 \qquad \text{(A.1b)} \qquad (2)$$

$$\vec{\nabla}\times\vec{E}(\vec{r},t) = -\frac{\partial}{\partial t}\vec{B}(\vec{r},t) \qquad \text{(A.1c)} \qquad (3)$$

$$\vec{\nabla}\times\vec{B}(\vec{r},t) = \frac{1}{c^2}\frac{\partial}{\partial t}\vec{E}(\vec{r},t) + \frac{1}{\varepsilon_0 c^2}\,\vec{j}(\vec{r},t) \qquad \text{(A.1d)} \qquad (4)$$

## 1.2 Continuity equation

From $\varepsilon_0\,\partial_t$ (A.1a), using $\partial_x\partial_t = \partial_t\partial_x$,

$$\frac{\partial}{\partial t}\rho(\vec{r},t) = \varepsilon_0\,\vec{\nabla}\cdot\big(\partial_t\vec{E}(\vec{r},t)\big), \qquad (5)$$

and from $\varepsilon_0 c^2\,\vec{\nabla}\cdot$ (A.1d),

$$\begin{aligned}\vec{\nabla}\cdot\vec{j}(\vec{r},t) &= \vec{\nabla}\cdot\big(\vec{\nabla}\times\vec{B}(\vec{r},t)\big) - \varepsilon_0\,\vec{\nabla}\cdot\big(\partial_t\vec{E}(\vec{r},t)\big)\\ &\overset{\mathrm{div}(\mathrm{rot}\,\vec{F})=0}{=} -\varepsilon_0\,\vec{\nabla}\cdot\big(\partial_t\vec{E}(\vec{r},t)\big). \qquad (6)\end{aligned}$$

Adding the two gives the continuity equation

$$\frac{\partial}{\partial t}\rho(\vec{r},t) + \vec{\nabla}\cdot\vec{j}(\vec{r},t) = 0. \qquad \text{(A.3)}$$

## 1.3 Point-particle solutions

Solutions of the continuity equation:

$$\rho(\vec{r},t) = \sum_\alpha q_\alpha\,\delta\big(\vec{r}-\vec{r}_\alpha(t)\big) \qquad \text{(A.5a)} \qquad (7)$$

$$\vec{j}(\vec{r},t) = \sum_\alpha q_\alpha\,\vec{v}_\alpha(t)\,\delta\big(\vec{r}-\vec{r}_\alpha(t)\big) \qquad \text{(A.5b)} \qquad (8)$$

*Proof.* Note that

$$\begin{aligned}\partial_t\,\delta\big(\vec{r}-\vec{r}_\alpha(t)\big) &= \partial_t\Big(\delta\big(r_x - r_x^\alpha(t)\big)\delta\big(r_y - r_y^\alpha(t)\big)\delta\big(r_z - r_z^\alpha(t)\big)\Big)\\ &\overset{\text{chain rule}}{=} -\sum_{i=x,y,z}\dot{r}_i^\alpha(t)\,\partial_{r_i}\delta\big(r_i - r_i^\alpha(t)\big) = -\dot{\vec{r}}_\alpha(t)\cdot\vec{\nabla}\,\delta\big(\vec{r}-\vec{r}_\alpha(t)\big), \qquad (9)\end{aligned}$$

and

$$\vec{\nabla} \cdot \Big(\vec{v}_\alpha(t)\,\delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big) = \sum_{i=x,y,z} \partial_{r_i} \Big(v_i^\alpha(t)\,\delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big)$$
$$= \sum_{i=x,y,z} v_i^\alpha(t)\,\partial_{r_i}\delta\big(\vec{r} - \vec{r}_\alpha(t)\big) = \vec{v}_\alpha(t) \cdot \vec{\nabla}\,\delta\big(\vec{r} - \vec{r}_\alpha(t)\big). \tag{10}$$

Therefore

$$\frac{\partial}{\partial t}\rho(\vec{r}, t) = -\sum_\alpha q_\alpha\,\dot{\vec{r}}_\alpha(t) \cdot \Big(\vec{\nabla}\,\delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big) = -\vec{\nabla} \cdot \vec{j}(\vec{r}, t), \tag{11}$$

which is (A.3). □

Note that these solutions are not unique (one equation for four components). Intermediate result: the current can be defined by “charge times velocity”.